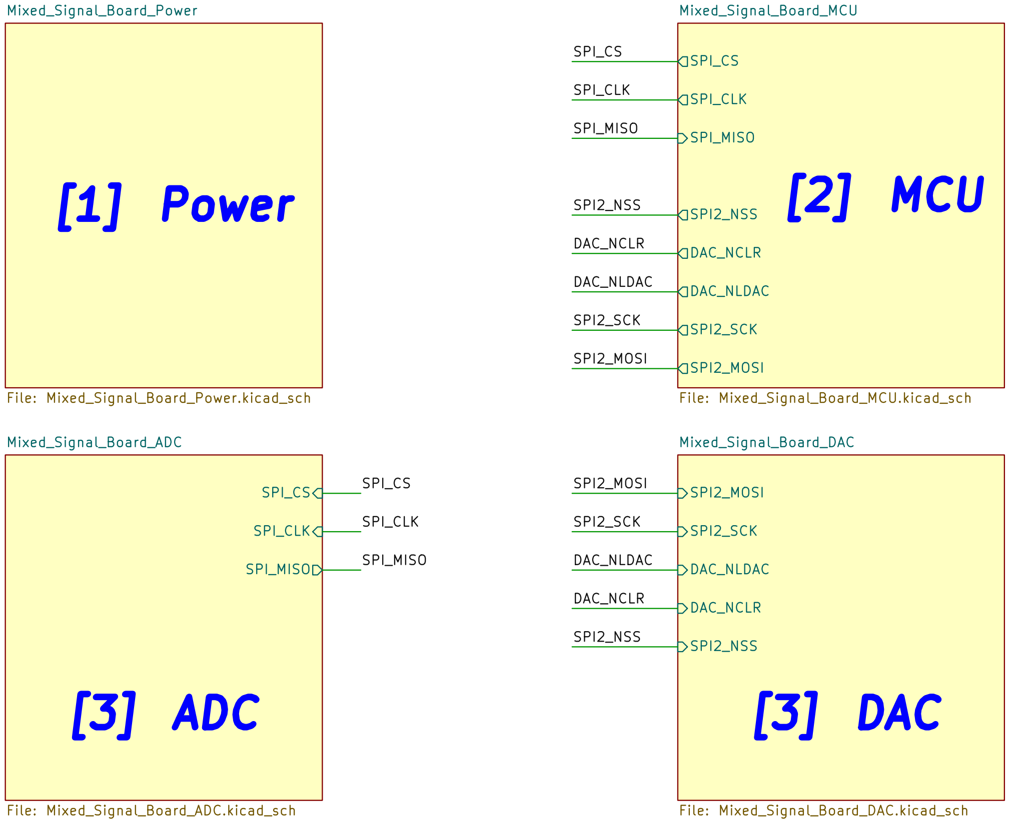
Mixed Signal Board

Hardware Design for Analog Signal Acquisition and Generation Using ADC and DAC



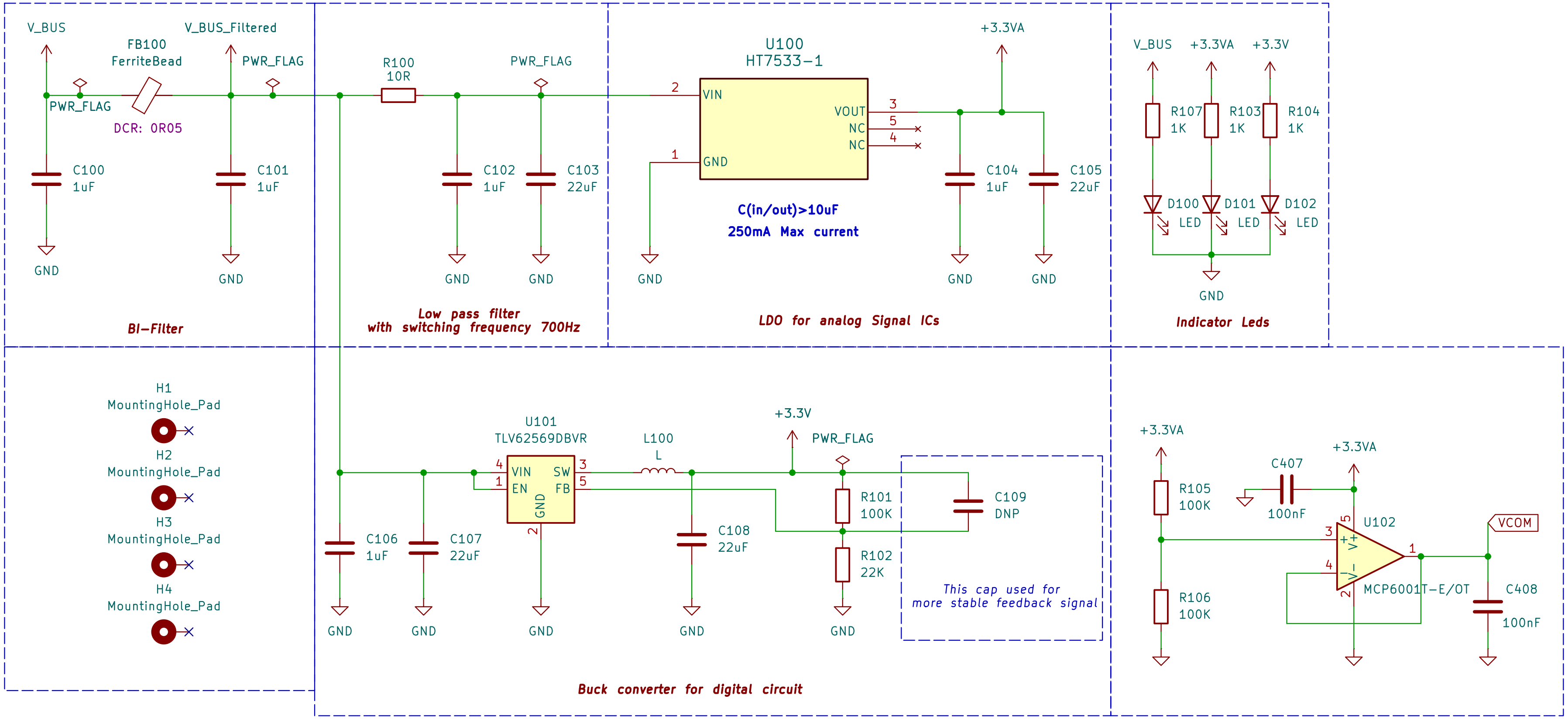
Sheet: /
File: Mixed_Signal_Board_FEDEVEL_Course.kicad_sch

Title: Mixed_Signal_Board

Size: A4 | Date: 2026-03-29 | **Rev: A**

KiCad E.D.A. 10.0.1 | Id: 1/5

[1] Power



Buck design notes

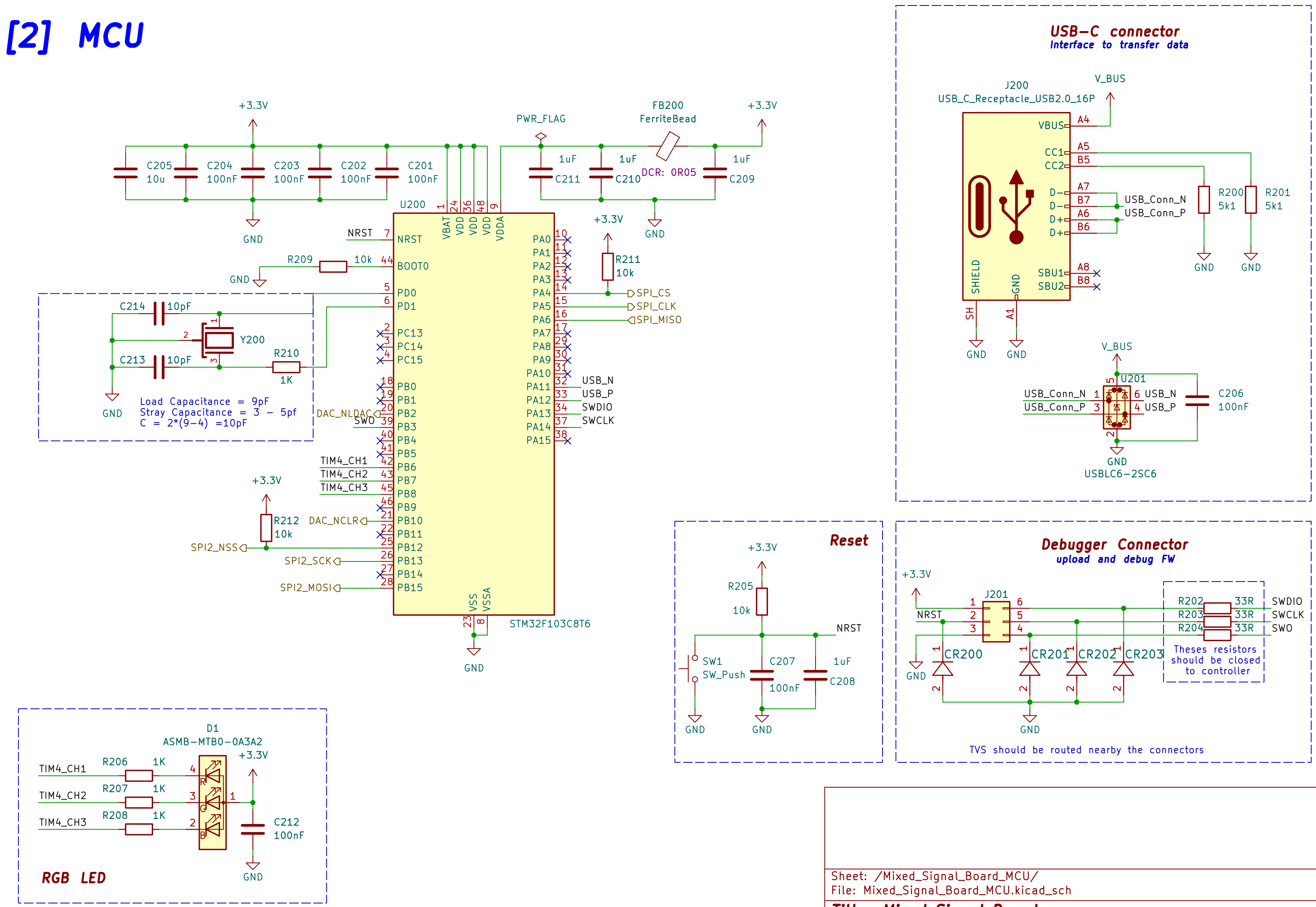
V(out) = 0.6V*(1+(R1/R)) 3.327V = 0.6*(1+(100K/22K)) R1 -> upper feedback resistor R2 -> lower feedback resistor	I_load max = 250mA F_switching = 1.5MHz
Recommended value of inductor regarding datasheet between 2.2uH and 1uH for lightloads smaller inductors more suitable to avoid operating in Discontinuous Conduction Mode	

Sheet: /Mixed_Signal_Board_Power/
File: Mixed_Signal_Board_Power.kicad_sch

Title: Mixed_Signal_Board

Size: A4 Date: 2026-03-29 **Rev: A**
KiCad E.D.A. 10.0.1 Id: 2/5

[2] MCU

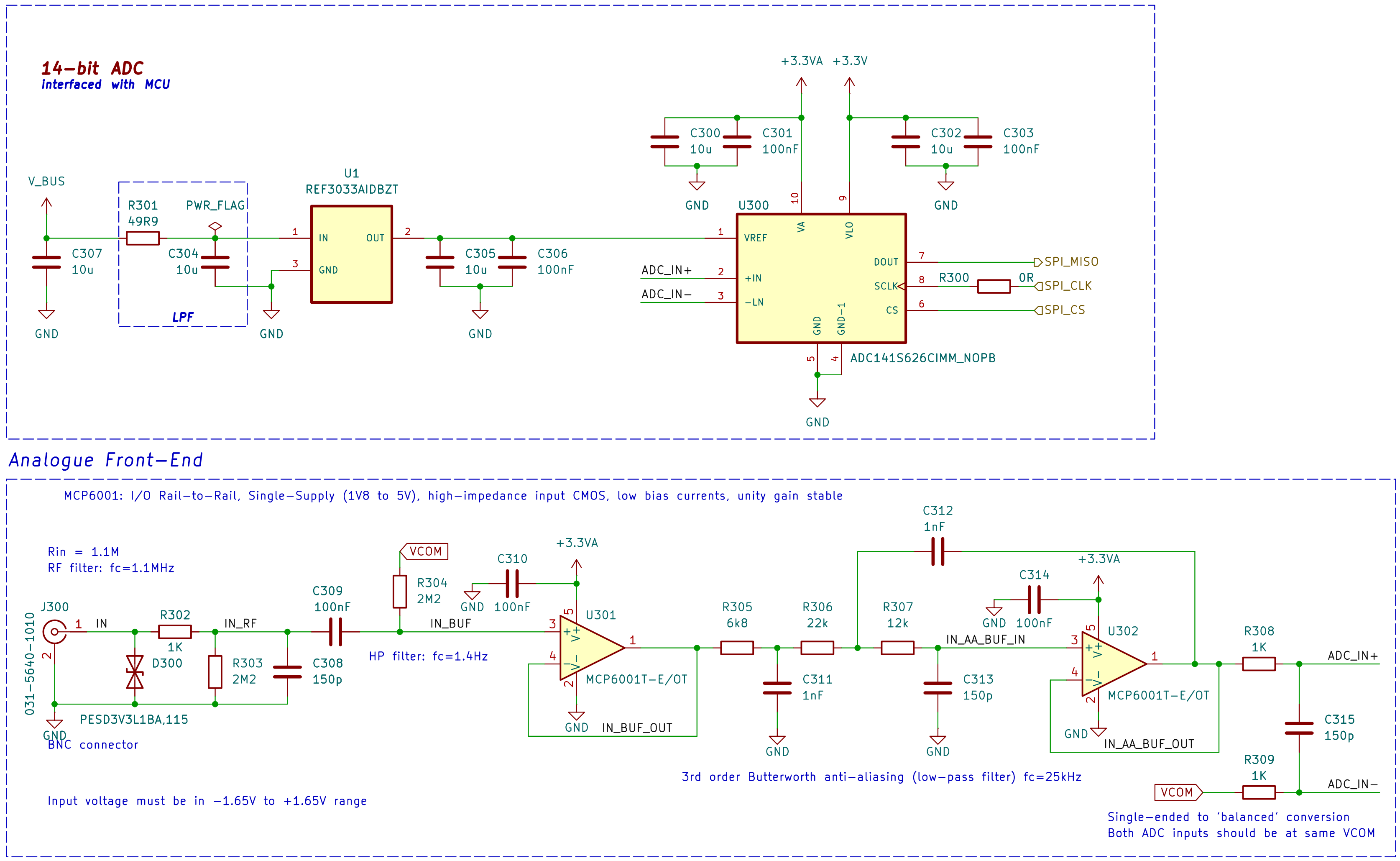


Sheet: /Mixed_Signal_Board_MCU/
File: Mixed_Signal_Board_MCU.kicad_sch

Title: Mixed_Signal_Board

Size: A4	Date: 2026-04-18	Rev: A
KiCad E.D.A. 10.0.1		Id: 3/5

[3] ADC

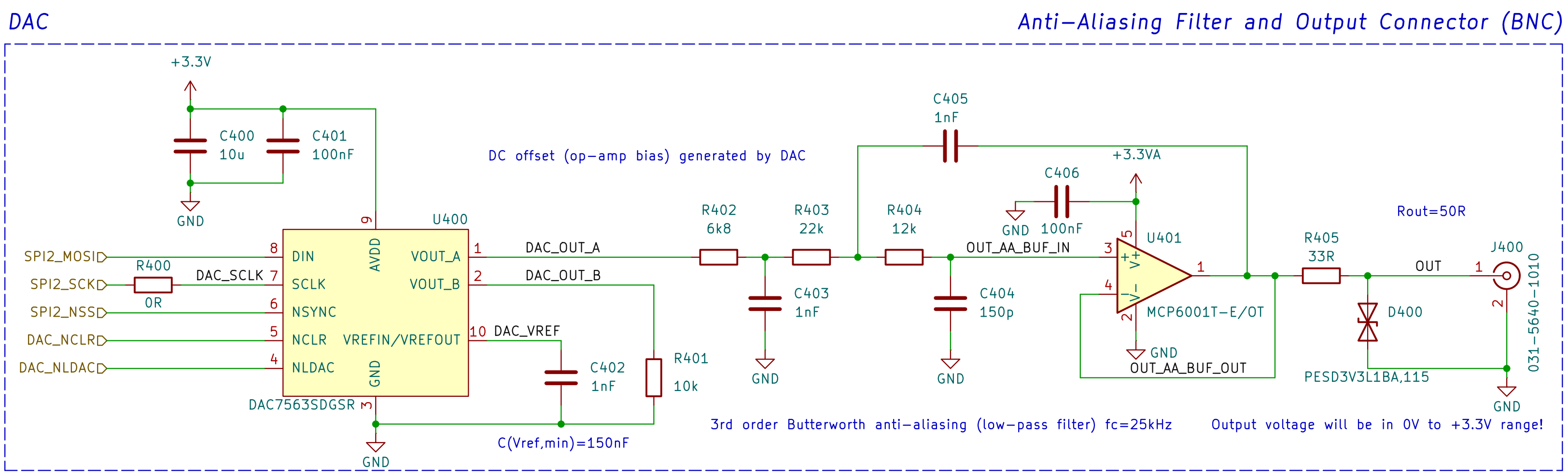


Sheet: /Mixed_Signal_Board_ADC/
File: Mixed_Signal_Board_ADC.kicad_sch

Title: Mixed_Signal_Board

Size: A4 Date: 2026-05-02 **Rev: A**
KiCad E.D.A. 10.0.1 Id: 4/5

[4] DAC & Analogue Output Circuitry



Sheet: /Mixed_Signal_Board_DAC/
File: Mixed_Signal_Board_DAC.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.1		Id: 5/5